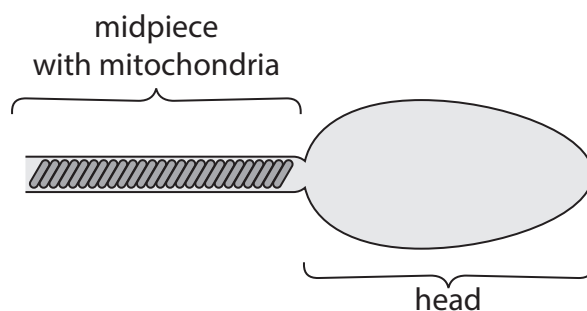


Answer ALL questions.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 A sperm cell is specialised for its function.

(a) The diagram shows part of a labelled sperm cell.



(i) Complete the diagram by drawing and labelling a flagellum and an acrosome.

(1)

(ii) Give the function of the mitochondria in the midpiece of the sperm.

(1)

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(b) Some male mice will mate with only one female and some will mate with more than one female.

The swimming speed of sperm cells of mice has been investigated.

The graph shows the relationship between midpiece length and swimming speed of sperm cells of mice.



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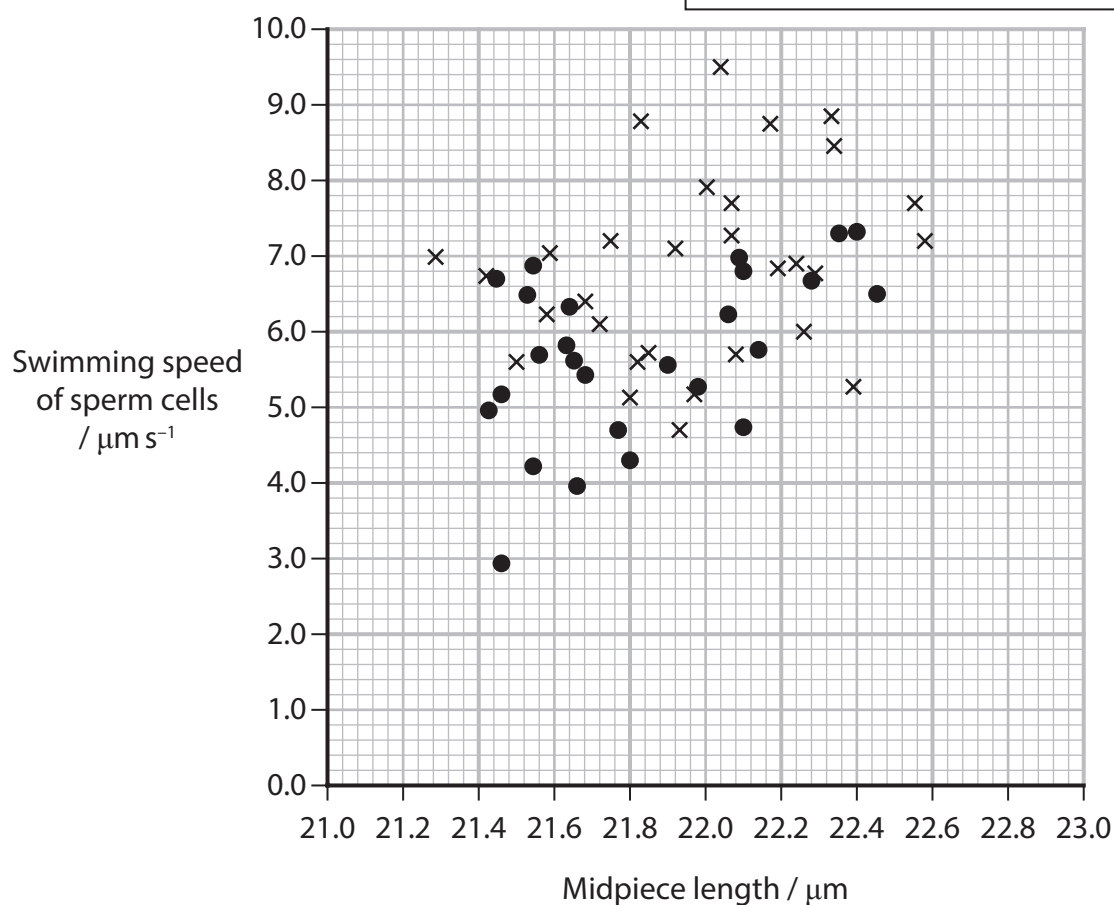
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Key:

● male mates with only one female

× male mates with more than one female



- (i) Give **two** conclusions that can be drawn from these data.

(2)

1

.....

.....

2

.....

.....

- (ii) Give the name of a statistical test that could be used to assess the strength of the relationship between length of midpiece and swimming speed of sperm cells.

(1)

.....

(Total for Question 1 = 5 marks)